Machine Learning Project 1

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Results

Multinomial Naive Bayes

Dataset	Accuracy	Precision	Recall	F1-Score
enron1	0.9364	0.8947	0.9128	0.9037
enron2	0.9435	0.8503	0.9615	0.9025
enron4	0.9779	0.9773	0.9923	0.9848

Bernoulli Naive Bayes

Dataset	Accuracy	Precision	Recall	F1-Score
enron1	0.8355	0.8627	0.5906	0.7012
enron2	0.8473	0.8132	0.5692	0.6697
enron4	0.9613	0.9490	1.0000	0.9738

Logistic Regression

Dataset	Representation	Lamda	Accuracy	Precision	Recall	F1-Score
enron1	bow	10.0	0.9561	0.9108	0.9597	0.9346
enron1	bernoulli	1.0	0.9539	0.9051	0.9597	0.9316
enron2	bow	10.0	0.9310	0.9292	0.8077	0.8642
enron2	bernoulli	0.001	0.9644	0.9124	0.9615	0.9363
enron4	bow	0.001	0.9650	0.9537	1.0000	0.9763
enron4	bernoulli	0.001	0.9687	0.9583	1.0000	0.9787

Discussion

Hyperparameter tuning:

1) 70/30 split of training data (train/validation)
2) Tested lambda values: [0.001, 0.01, 0.1, 1.0, 10.0]
3) Selected lambda with best validation accuracy
4) Retrained on full training set with best lambda
5) Evaluated on test set

Q1: Did Naive Bayes or Logistic Regression perform better? Why?

On the majority of datasets and representations, Logistic Regression performed better overall than Naive Bayes. Logistic Regression continuously obtained higher F1-scores because it could learn discriminative decision boundaries instead of assuming feature independence, even if Multinomial Naive Bayes was competitive particularly on enron4.

Q2: Which combination of algorithm and data representation yielded the best performance? Why?

The best combination was Multinomial Naive Bayes with Bag of Words approach F1 = 0.9848 on enron4. It is mostly because independent assuming is less ineffective when a few tokens influence discrimination. The generative model was able to estimate class-conditional likelihoods more accurately than binary indicators thanks to the signal that counting information preserves. Logistic regression with Bernoulli was very close too (enron4 dataset F-1 score: 0.9787). It performed excellent as well as it has discriminative learning ability.

Q3: Did Multinomial Naive Bayes perform better than Logistic Regression on the Bag of Words representation? Explain.

No, Multinomial Naive Bayes (F1 = 0.9037 on Enron1, 0.9025 on Enron2) fared worse than Logistic Regression with Bag of Words (F1 = 0.9763 on Enron4, 0.9346 on Enron1). Even though Multinomial Naive Bayes does a good job of modeling raw word counts, Logistic Regression performs better because it can handle word correlations and weigh features more efficiently.

Q4: Did Bernoulli Naive Bayes perform better than Logistic Regression on the Bernoulli representation? Explain.

No. Bernoulli Naive Bayes was routinely outperformed by Logistic Regression on Bernoulli representation. Because Bernoulli Naive Bayes relies on binary likelihoods, which are less flexible, and presupposes high feature independence, it had trouble. In contrast, logistic regression improves prediction performance by identifying dependencies and learning the best feature weights.

4. Conclusion

Across all datasets, logistic regression performs well, particularly when used with the Bernoulli representation. Despite being quick and easy, naive Bayes techniques are less competitive because to their independence assumptions.